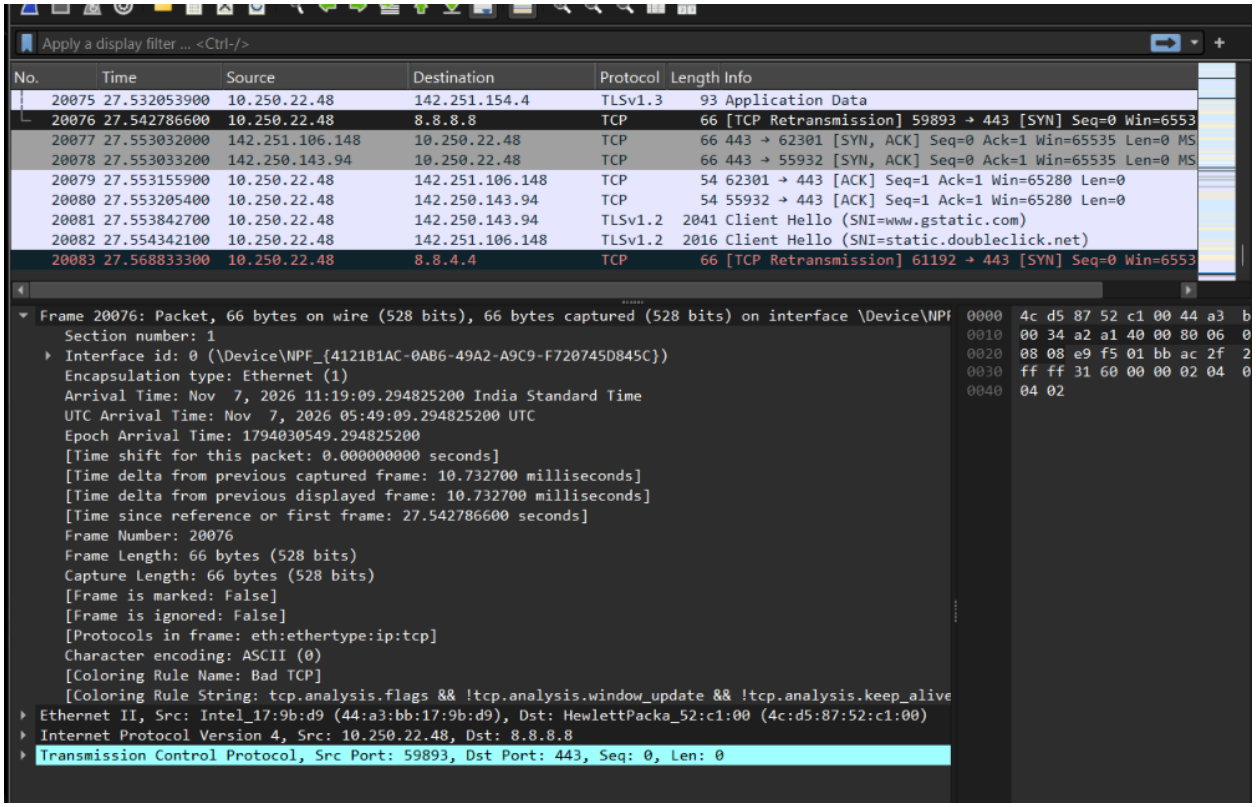
CNNP SKILL-1

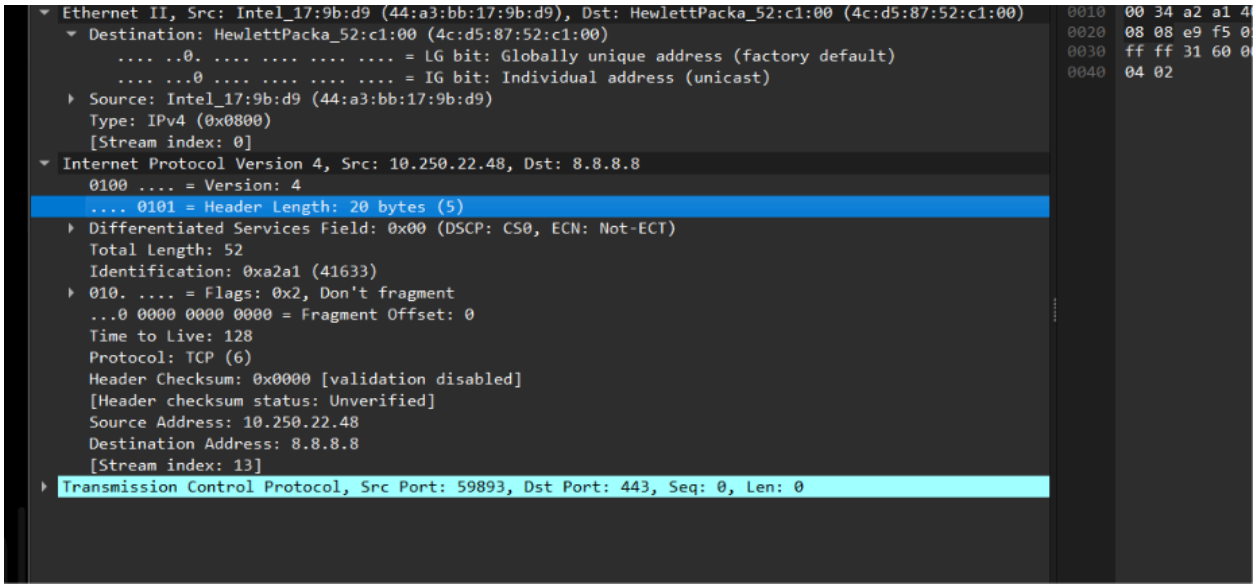
Praneeth Kumar Gadila
2520030257

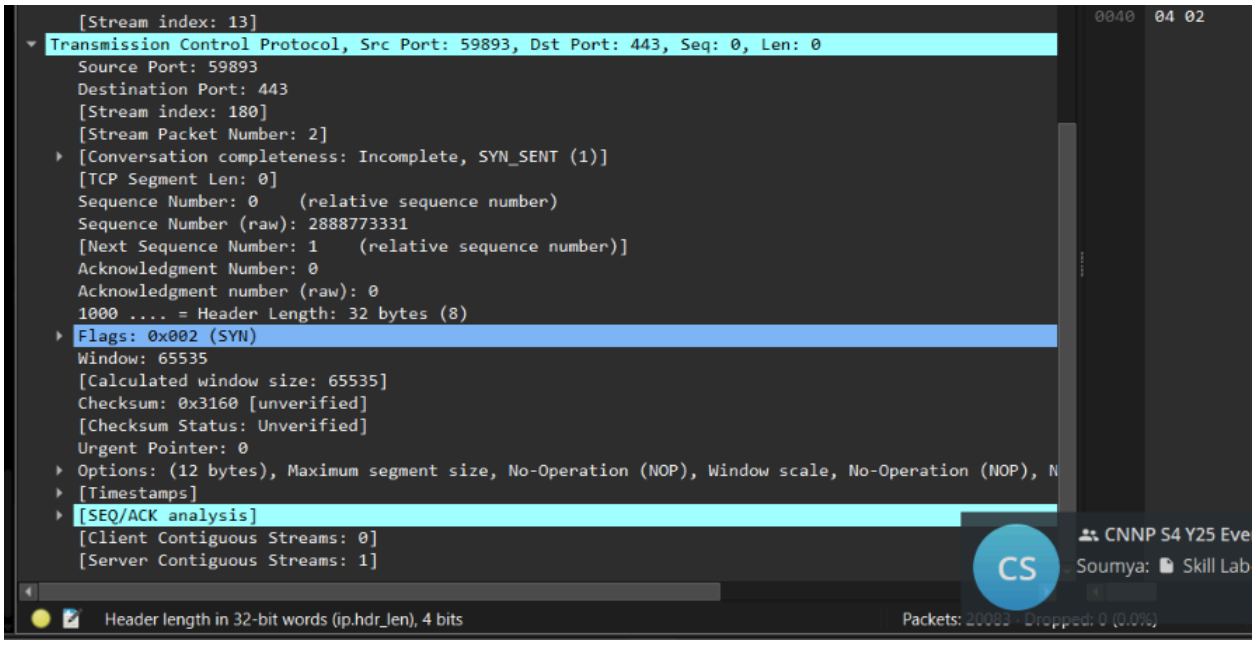
Encapsulation Builder: Analyze the overhead introduced by protocol headers and determine how encapsulation affects network efficiency.

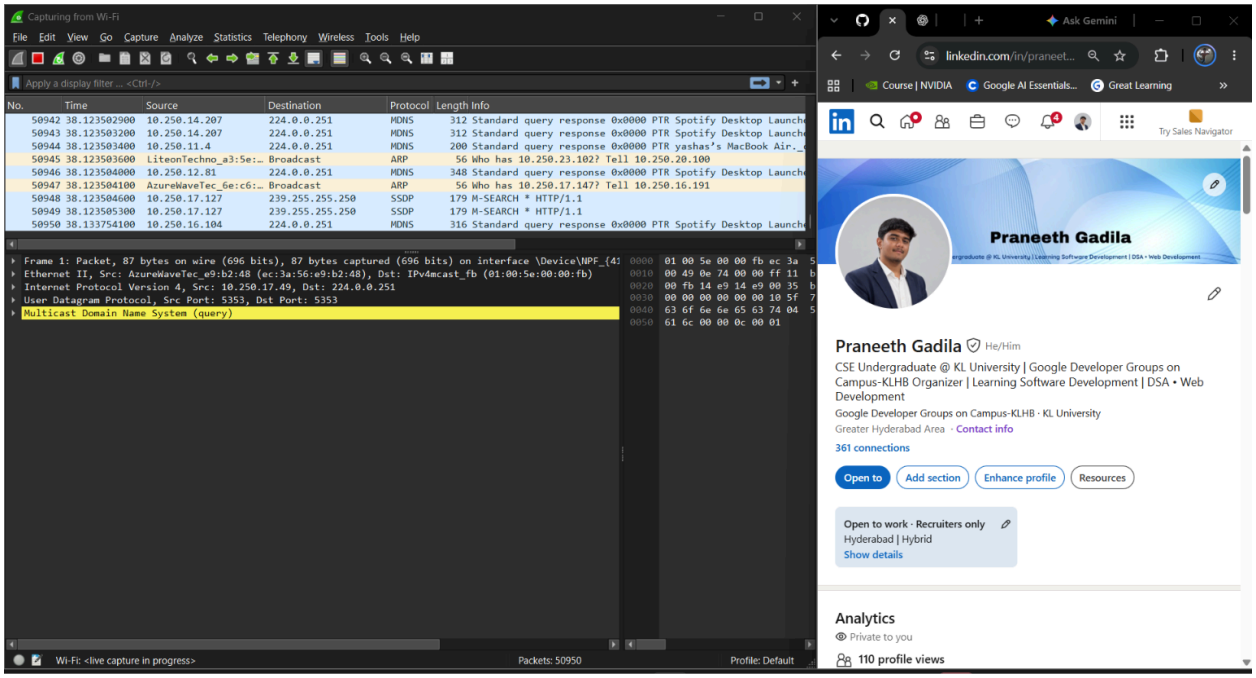
Aim: To capture network packets and understand how Ethernet, IP, and TCP/UDP headers are added during data transmission. Whenever data travels through a computer network, each protocol layer adds its own header to the data. These headers introduce overhead.

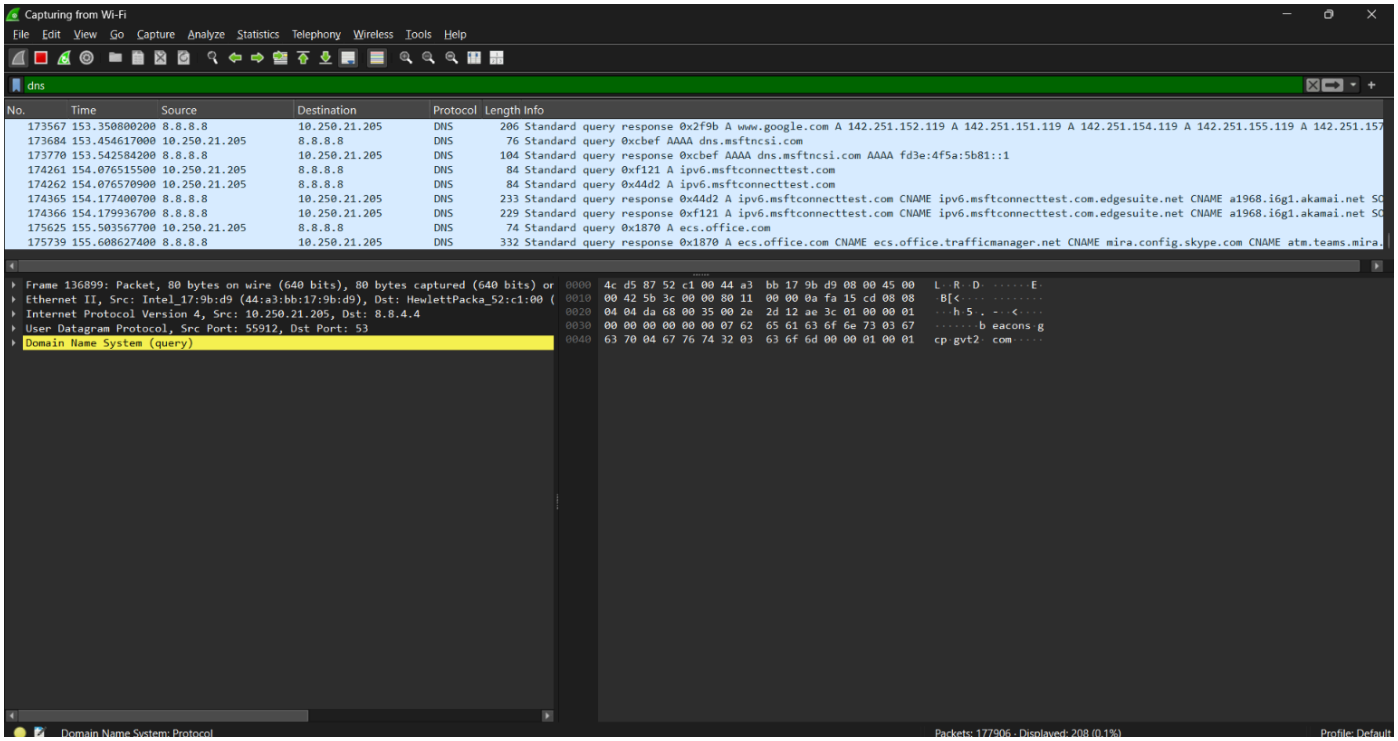
Using Wireshark, we will capture real network packets, examine Ethernet, IP and TCP/UDP headers, calculate the total overhead, and determine how efficiently the network carries application data.

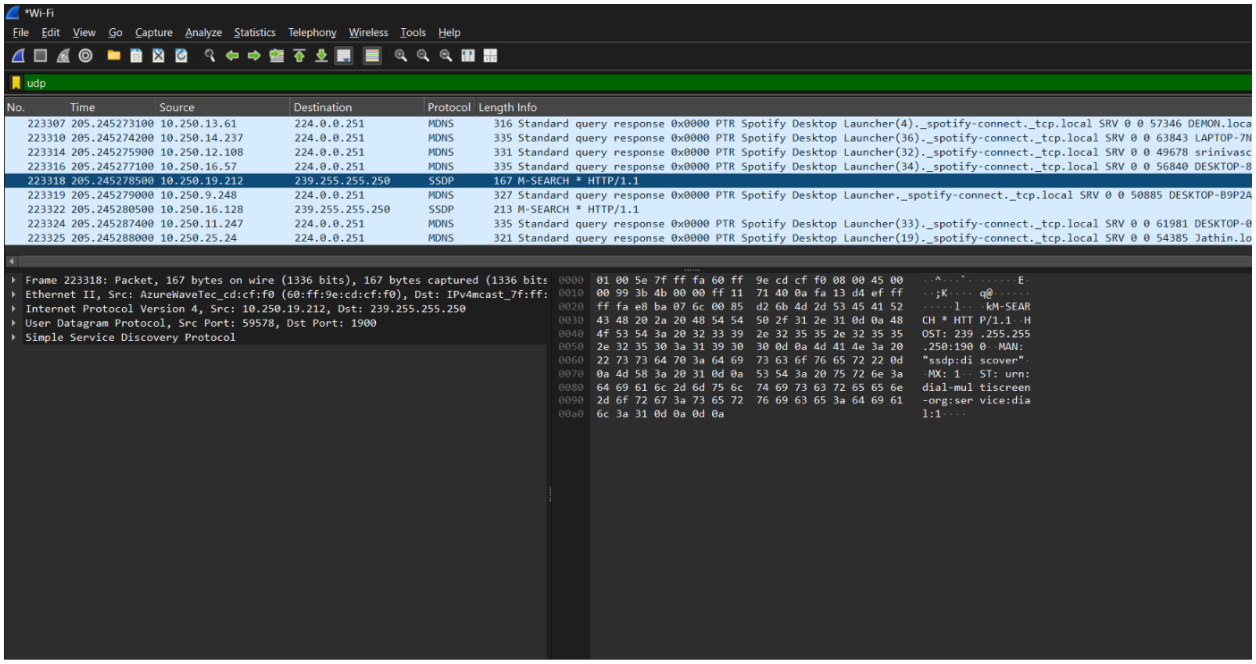



Ethernet II, Src: Intel_17:9b:d9 (44:a3:bb:17:9b:d9), Dst: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
Destination: HewlettPacka_52:c1:00 (4c:d5:87:52:c1:00)
.... ..0. .... = LG bit: Globally unique address (factory default)
....0 .... = IG bit: Individual address (unicast)
Source: Intel_17:9b:d9 (44:a3:bb:17:9b:d9)
Type: IPv4 (0x0800)
[Stream index: 0]
Internet Protocol Version 4, Src: 10.250.22.48, Dst: 8.8.8.8
0100 .... = Version: 4
.... 0101 = Header Length: 20 bytes (5)
Differentiated Services Field: 0x00 (DSCP: CS0, ECN: Not-ECT)
Total Length: 52
Identification: 0xa2a1 (41633)
010. .... = Flags: 0x2, Don't fragment
...0 0000 0000 0000 = Fragment Offset: 0
Time to Live: 128
Protocol: TCP (6)
Header Checksum: 0x0000 [validation disabled]
[Header checksum status: Unverified]
Source Address: 10.250.22.48
Destination Address: 8.8.8.8
[Stream index: 13]
Transmission Control Protocol, Src Port: 59893, Dst Port: 443, Seq: 0, Len: 0
0010 00 34 a2 a1 4
0020 08 08 e9 f5 0
0030 ff ff 31 60 0
0040 04 02


[Stream index: 13]
Transmission Control Protocol, Src Port: 59893, Dst Port: 443, Seq: 0, Len: 0
Source Port: 59893
Destination Port: 443
[Stream index: 180]
[Stream Packet Number: 2]
[Conversation completeness: Incomplete, SYN_SENT (1)]
[TCP Segment Len: 0]
Sequence Number: 0 (relative sequence number)
Sequence Number (raw): 2888773331
[Next Sequence Number: 1 (relative sequence number)]
Acknowledgment Number: 0
Acknowledgment number (raw): 0
1000 .... = Header Length: 32 bytes (8)
Flags: 0x002 (SYN)
Window: 65535
[Calculated window size: 65535]
Checksum: 0x3160 [unverified]
[Checksum Status: Unverified]
Urgent Pointer: 0
Options: (12 bytes), Maximum segment size, No-Operation (NOP), Window scale, No-Operation (NOP), N
[Timestamps]
[SEQ/ACK analysis]
[Client Contiguous Streams: 0]
[Server Contiguous Streams: 1]
0040 04 02
CS
CNNP S4 Y25 Eve
Soumya: Skill Lab
Header length in 32-bit words (ip.hdr_len), 4 bits
Packets: 20093 · Dropped: 0 (0.0%)


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
Apply a display filter ... <Ctrl-/>
No. Time Source Destination Protocol Length Info
50942 38.123502900 10.250.14.207 224.0.0.251 MDNS 312 Standard query response 0x0000 PTR Spotify Desktop Launch
50943 38.123503200 10.250.14.207 224.0.0.251 MDNS 312 Standard query response 0x0000 PTR Spotify Desktop Launch
50944 38.123503400 10.250.11.4 224.0.0.251 MDNS 200 Standard query response 0x0000 PTR yashas's MacBook Air._
50945 38.123503600 LiteonTechno_a3:5e:... Broadcast ARP 56 Who has 10.250.23.102? Tell 10.250.20.100
50946 38.123504000 10.250.12.81 224.0.0.251 MDNS 348 Standard query response 0x0000 PTR Spotify Desktop Launch
50947 38.123504100 AzureWaveTec_6e:c6:... Broadcast ARP 56 Who has 10.250.17.147? Tell 10.250.16.191
50948 38.123504600 10.250.17.127 239.255.255.250 SSDP 179 M-SEARCH * HTTP/1.1
50949 38.123505300 10.250.17.127 239.255.255.250 SSDP 179 M-SEARCH * HTTP/1.1
50950 38.133754100 10.250.16.104 224.0.0.251 MDNS 316 Standard query response 0x0000 PTR Spotify Desktop Launch
Frame 1: Packet, 87 bytes on wire (696 bits), 87 bytes captured (696 bits) on interface \Device\NPF_{4
Ethernet II, Src: AzureWaveTec_e9:b2:48 (ec:3a:56:e9:b2:48), Dst: IPv4mcast_fb (01:00:5e:00:00:fb)
Internet Protocol Version 4, Src: 10.250.17.49, Dst: 224.0.0.251
User Datagram Protocol, Src Port: 5353, Dst Port: 5353
Multicast Domain Name System (query)
0000 01 00 5e 00 00 fb ec 3a
0010 00 49 0e 74 00 00 ff 11
0020 00 fb 14 e9 14 e9 00 35
0030 00 00 00 00 00 00 10 5f
0040 63 6f 6e 6e 65 63 74 04
0050 61 6c 00 00 0c 00 01
Wi-Fi: <live capture in progress>
Packets: 50950
Profile: Default
linkedin.com/in/praneet...
Ask Gemini
Course | NVIDIA
Google AI Essentials...
Great Learning
Try Sales Navigator
Praneeth Gadila
Praneeth Gadila He/Him
CSE Undergraduate @ KL University | Google Developer Groups on Campus-KLHB Organizer | Learning Software Development | DSA • Web Development
Google Developer Groups on Campus-KLHB · KL University
Greater Hyderabad Area · Contact info
361 connections
Open to
Add section
Enhance profile
Resources
Open to work · Recruiters only
Hyderabad | Hybrid
Show details
Analytics
Private to you
110 profile views


Capturing from Wi-Fi
File Edit View Go Capture Analyze Statistics Telephony Wireless Tools Help
dns
No. Time Source Destination Protocol Length Info
173567 153.350800200 8.8.8.8 10.250.21.205 DNS 206 Standard query response 0x2f9b A www.google.com A 142.251.152.119 A 142.251.151.119 A 142.251.154.119 A 142.251.155.119 A 142.251.157
173684 153.454617000 10.250.21.205 8.8.8.8 DNS 76 Standard query 0xcbef AAAA dns.msftncsi.com
173770 153.542584200 8.8.8.8 10.250.21.205 DNS 104 Standard query response 0xcbef AAAA dns.msftncsi.com AAAA fd3e:4f5a:5b81::1
174261 154.076515500 10.250.21.205 8.8.8.8 DNS 84 Standard query 0xf121 A ipv6.msftconnecttest.com
174262 154.076570900 10.250.21.205 8.8.8.8 DNS 84 Standard query 0x44d2 A ipv6.msftconnecttest.com
174365 154.177400700 8.8.8.8 10.250.21.205 DNS 233 Standard query response 0x44d2 A ipv6.msftconnecttest.com CNAME ipv6.msftconnecttest.com.edgesuite.net CNAME a1968.i6g1.akamai.net SO
174366 154.179936700 8.8.8.8 10.250.21.205 DNS 229 Standard query response 0xf121 A ipv6.msftconnecttest.com CNAME ipv6.msftconnecttest.com.edgesuite.net CNAME a1968.i6g1.akamai.net SO
175625 155.503567700 10.250.21.205 8.8.8.8 DNS 74 Standard query 0x1870 A ecs.office.com
175739 155.608627400 8.8.8.8 10.250.21.205 DNS 332 Standard query response 0x1870 A ecs.office.com CNAME ecs.office.trafficmanager.net CNAME mira.config.skype.com CNAME atm.teams.mira.
Frame 136899: Packet, 80 bytes on wire (640 bits), 80 bytes captured (640 bits) or
Ethernet II, Src: Intel_17:9b:d9 (44:a3:bb:17:9b:d9), Dst: HewlettPacka_52:c1:00 (
Internet Protocol Version 4, Src: 10.250.21.205, Dst: 8.8.4.4
User Datagram Protocol, Src Port: 55912, Dst Port: 53
Domain Name System (query)
0000 4c d5 87 52 c1 00 44 a3 bb 17 9b d9 08 00 45 00
0010 00 42 5b 3c 00 00 80 11 00 00 0a fa 15 cd 08 08
0020 04 04 da 68 00 35 00 2e 2d 12 ae 3c 01 00 00 01
0030 00 00 00 00 00 00 07 62 65 61 63 6f 6e 73 03 67
0040 63 70 04 67 76 74 32 03 63 6f 6d 00 00 01 00 01
Domain Name System: Protocol
Packets: 177906 · Displayed: 208 (0.1%)
Profile: Default



output-

Parameter	TCP Packet (bytes)	UDP Packet (bytes)
Frame length	66	78
Ethernet header length	14	14
Ip header length	20	20
Transport header length	32	8
Ip total length	52	64
Transport header length	32	8
payload	0	56